

Momentum in Large-Cap Equities: Does the Classic 12–1 Strategy Still Work?

A Survivorship-Bias-Free Examination of the S&P 500, 2006–2024

Darshan Sathish Kumar*

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Abstract

We evaluate the canonical 12–1 month cross-sectional momentum strategy applied to the S&P 500 over the period March 2006–December 2024, using a survivorship-bias-free constituent history, gap-aware return construction, and a granular turnover-based transaction cost model. Each month we rank active constituents by their prior 12-month return (skipping the most recent month), go long the top decile and short the bottom decile equal-weighted, and rebalance monthly. The strategy produces a net annualized return of -2.79% , annualized volatility of 20.66% , a Sharpe ratio of -0.23 , and a maximum drawdown of -81% at 10 basis points per side. Leg decomposition reveals that the signal is economically valid—the long leg earns $+7.9\%$ annualized and the short leg loses -9.1% —but systematic shorting of past losers proves catastrophically unprofitable during the 2009 and 2020 momentum crash episodes, when deeply distressed stocks recovered by 40–100% within single calendar months. Average monthly one-way turnover is 53.6%; transaction costs account for only $\approx 0.05\%$ per month and are not the primary driver of losses. Factor regressions (CAPM through FF5+UMD, with Newey–West errors) show that R^2 rises from 0.15 under CAPM to 0.82 once UMD is included, with a statistically significant negative alpha of -4.0% annually ($t = -3.23$). These results confirm that naïve large-cap momentum is primarily a leveraged exposure to the well-documented UMD factor, with no residual alpha once crash risk and frictions are accounted for, and that the strategy’s poor performance over this period is driven by the short leg rather than by any failure of the ranking signal itself.

Keywords: momentum; cross-sectional strategies; momentum crashes; survivorship bias; transaction costs; Fama–French factors; S&P 500.

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1 Introduction

The momentum anomaly—the empirical regularity that stocks with strong recent returns tend to continue outperforming those with weak recent returns—has occupied a central place in the asset

*Email: darshansathishkumar@gmail.com. Code and replication data: <https://github.com/dshan12/Momentum-Research>.

pricing literature since Jegadeesh and Titman (1993) first documented it systematically in U.S. equities. Three decades on, momentum remains simultaneously one of the most replicated and most debated findings in finance. It appears across asset classes (Asness et al., 2013), geographies (Rouwenhorst, 1998), and time periods stretching back to Victorian-era stock markets (Geczy and Samonov, 2013). Yet it is also a strategy that has spectacularly failed practitioners at precisely the worst moments (Daniel and Moskowitz, 2016), and whose theoretical foundations remain contested (Barberis et al., 1998; Hong and Stein, 1999).

The U.S. large-cap equity universe—specifically the S&P 500—presents a particularly challenging testing ground. Constituent stocks are among the most liquid, most widely followed, and most efficiently priced securities in the world. Any systematic pattern in their returns attracts immediate arbitrage capital, compressing risk premia rapidly. Moreover, the post-2000 era has been unusually hostile to momentum strategies: two of the most severe momentum crashes on record—March–May 2009 and November 2020—occurred within this window, and the prolonged bull market of 2010–2024 created conditions under which past losers persistently recovered, undermining the short leg of momentum portfolios.

Research question. *Can a naïve 12–1 month momentum rule, applied to S&P 500 constituents with survivorship-bias-free data and realistic transaction costs, generate statistically significant abnormal returns over the 2006–2024 period? And if not, which component of the strategy—the long leg, the short leg, or transaction costs—is responsible for its failure?*

Contribution. This paper makes four contributions relative to prior replication work. First, we reconstruct a dated S&P 500 membership history from Wikipedia change logs, enabling gap-aware return computation that avoids both survivorship bias (including only current constituents) and re-entry bias (computing returns only across consecutive months in which a stock is an active constituent). Second, we implement a granular turnover-based cost model that tracks mark-to-market weight drift between rebalancing dates, producing realistic one-way turnover estimates of $\approx 54\%$ per month. Third, we decompose gross P&L into long-leg and short-leg contributions, providing a clearer diagnosis of where the strategy fails than aggregate performance statistics alone. Fourth, we contextualize our results against the Fama–French UMD factor over the same sample period, demonstrating that the strategy’s return profile closely tracks UMD—including its crash behavior—while underperforming it net of costs.

Preview of results. The signal works as a cross-sectional ranking mechanism: the long leg earns +7.9% annualized and the short leg loses −9.1% in absolute terms, confirming that past winners outperform past losers on average. However, the combined strategy is unprofitable because the short leg bears catastrophic losses during two momentum crash episodes that together erase more than half the portfolio’s value. Transaction costs, while non-trivial at $\approx 0.65\%$ annualized, are a secondary concern. The strategy’s return variation is almost entirely spanned by the UMD factor ($R^2 = 0.82$), and its alpha is negative and significant once UMD is controlled for. These findings

reinforce the view that momentum in large-cap U.S. equities is a priced risk factor—specifically, crash risk—rather than a persistent source of exploitable alpha.

2 Related Literature

Jegadeesh and Titman (1993) establish the foundational result: strategies that buy past six- to twelve-month winners and sell past losers earn abnormal returns of approximately 1% per month over a three- to twelve-month holding period in U.S. equities. Carhart (1997) show that momentum largely explains the persistence of mutual fund performance, cementing it as a systematic risk factor. Fama and French (1996) incorporate momentum into their three-factor framework’s “biggest embarrassment,” acknowledging that their model fails to capture it; the formal UMD factor is later constructed to address this gap (Fama and French, 2015).

The theoretical literature offers two broad classes of explanations. Behavioral models attribute momentum to investor underreaction (Barberis et al., 1998) or gradual information diffusion (Hong and Stein, 1999): prices adjust too slowly to fundamental news, creating predictable return continuation. Risk-based explanations posit that momentum portfolios earn compensation for bearing some form of systematic risk, though identifying the precise risk factor has proven elusive (Liu and Zhang, 2008; Johnson, 2002).

A second strand of literature documents the limits and costs of momentum strategies. Novy-Marx and Velikov (2016) demonstrate that momentum strategies require substantially higher turnover than previously appreciated and that realistic transaction costs—particularly for institutional investors trading at scale—can eliminate much of the gross alpha. Frazzini et al. (2015) estimate effective trading costs for a large institutional manager and find that momentum remains profitable after costs only when carefully implemented with patient execution. Lesmond et al. (2004) show that much of the gross momentum premium accrues in illiquid stocks where bid-ask spreads are prohibitive, calling into question the economic significance of the anomaly for large portfolios.

Perhaps most relevant to our findings is the literature on momentum crashes. Daniel and Moskowitz (2016) demonstrate that momentum strategies have highly negatively skewed return distributions, with occasional catastrophic losses (“crashes”) that occur during market rebounds following sharp downturns. The mechanism is straightforward: momentum strategies accumulate large short positions in stocks that have fallen dramatically; when the market reverses, these stocks exhibit the highest betas and recover most violently, inflicting concentrated losses on the short book. The authors document monthly losses of -49% (March–May 1932), -49% (March–May 2009), and other severe episodes. Barroso and Santa-Clara (2015) show that volatility-scaling the momentum position—reducing exposure when recent strategy volatility is elevated—substantially improves risk-adjusted performance and Sharpe ratios by approximately 50%.

More recent work examines the performance of momentum in the post-financial-crisis period. Asness (2019) document that momentum has delivered positive returns over very long samples (1926–present) but with significant time-variation and a notable deterioration post-2000 in large-cap

equity markets. Gulen et al. (2021) argue that increased arbitrage activity and factor crowding have compressed the momentum premium in large-cap stocks specifically, while it remains robust in smaller-cap universes.

Our paper contributes to this literature by providing a clean, reproducible implementation with full code transparency, a focus on the S&P 500 universe over a recent and economically challenging period, and an explicit decomposition that isolates the short leg as the source of losses rather than transaction costs or signal failure.

3 Data

3.1 Universe and Membership History

The investment universe consists of all firms that appeared as S&P 500 constituents at any point between January 2005 and December 2024. We identify 826 unique tickers over this window. Historical additions and deletions are reconstructed from the Wikipedia S&P 500 constituent change log, which records the date, added ticker, and removed ticker for each index rebalancing event since the 1990s. Starting from the current index composition, we roll backward through all change events to reconstruct the constituent set at each calendar month-end, then forward to produce a monthly membership panel spanning the full sample. Each ticker is marked as active only in months in which it was an S&P 500 constituent, ensuring that returns are computed only for stocks that were genuinely investable at the relevant time.

This construction eliminates survivorship bias in both directions: stocks deleted from the index (due to merger, bankruptcy, or downgrade) contribute their returns up to deletion and are then excluded, preventing the artificial inflation of historical performance from holding only survivors. The panel averages 393 active constituents per month (range: 0–502), with lower coverage in the early sample period as the Wikipedia history becomes less complete prior to 2005.

3.2 Prices and Return Construction

End-of-month adjusted closing prices are downloaded via `yfinance(interval="1mo", auto_adjust=True)`. All prices are snapped to month-end dates to ensure alignment with the membership panel.

Monthly returns are constructed as:

$$r_{i,t} = \frac{p_{i,t}}{p_{i,t-1}} - 1,$$

but only when *both* $p_{i,t}$ and $p_{i,t-1}$ are non-missing and the stock is an active constituent in month t . Critically, we do *not* use `pct_change` with gap-filling, which would compute the return from the last available price across an arbitrarily long gap. When a stock re-enters the index after an absence, its return in the re-entry month is set to missing, preventing the accumulation of a multi-month price move from being attributed to a single period. This is an important correction: without it,

stocks that crashed and recovered during an index absence can appear to have single-month returns of +100% or more, severely distorting cross-sectional rankings and portfolio returns.

A hard cap of $\pm 40\%$ is applied to monthly individual stock returns prior to winsorization. S&P 500 constituents essentially never move more than 40% in a single calendar month under normal market conditions; values beyond this threshold almost always reflect data artifacts (split adjustments, corporate actions) or represent returns on deeply distressed stocks that would not be replicable in a leveraged short book due to margin requirements and forced stop-outs.¹ Following the hard cap, returns are winsorized cross-sectionally at the 1st and 99th percentiles in months with at least 50 active names.

3.3 Factors and Benchmarks

Monthly Fama–French five-factor returns (Mkt-RF, SMB, HML, RMW, CMA), the risk-free rate (RF), and the momentum factor (UMD) are obtained from Kenneth French’s data library.² All factor returns are expressed as monthly decimals. The U.S. market benchmark is constructed as $\text{Mkt-RF} + R_f$, yielding +12.0% annualized over the 2006–2024 sample. The UMD factor produces +0.7% annualized over the same period, consistent with the documented deterioration of the momentum premium in large-cap U.S. equities post-2000.

3.4 Descriptive Statistics

Table 1 reports summary statistics for the net strategy returns and the U.S. market over the 226-month sample period. The strategy exhibits substantially higher volatility than the market (20.66% vs. 15.20% annualized), a large negative skew (−1.21 vs. −0.55), and excess kurtosis (5.84 vs. 1.40), reflecting the fat left tail generated by momentum crash months. The maximum drawdown of −81.2% dwarfs that of the market (−50.3%), driven by the 2009 crash from which the strategy never fully recovered before encountering the 2020 crash.

Table 1: Summary statistics. Monthly decimal returns; sample March 2006–December 2024 ($n = 226$ months). Sharpe computed with $r_f = 2\%$ annualized. $\text{Ann Vol} = \sigma_m \times \sqrt{12}$.

Series	n	Mean (m)	Sharpe	Ann Ret	Ann Vol	Skew	Kurt	Max DD
Strategy (net)	226	−0.0026	−0.23	−2.79%	20.66%	−1.21	5.84	−81.2%
US Market	226	0.0094	0.61	12.02%	15.20%	−0.55	1.40	−50.3%

¹Without this cap, multiple financial and cyclical stocks in our short book recorded monthly returns of +97%–+100% in April 2009, producing a single-month strategy loss of −53%. With the 40% cap, the April 2009 loss is −31%, which remains the worst month in the sample but is more consistent with the −49% cumulative loss documented for the UMD factor over March–May 2009 (Daniel and Moskowitz, 2016).

²https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

4 Methodology

4.1 Signal Construction

For each month t , the momentum signal for stock i is the cumulative log-price appreciation from $t-13$ to $t-1$:

$$\text{mom}_{i,t} = \ln P_{i,t-1} - \ln P_{i,t-13}.$$

This is the standard 12–1 specification: a 12-month lookback window that skips the most recent month. Skipping the most recent month is standard practice to avoid contamination by short-term price reversals (Jegadeesh, 1990), which would partially offset momentum at the one-month horizon. The signal is computed only for stocks with valid (non-missing) prices in both $t-1$ and $t-13$; stocks with gaps in either endpoint are excluded from the ranking in that month.

4.2 Portfolio Formation

At each month-end t , active constituents with valid momentum signals are ranked into percentiles. The long portfolio holds stocks in the top decile (≥ 90 th percentile) and the short portfolio holds stocks in the bottom decile (≤ 10 th percentile). Both legs are equal-weighted within the available names. We require a minimum of 20 stocks on each leg before executing; months failing this threshold are treated as no-trade months and excluded from the return series (25 of 252 calendar months are excluded on this basis). Weights formed at the end of month t are applied to returns realized during month $t+1$, ensuring no look-ahead bias.

4.3 Transaction Cost Model

Let $w_{i,t}$ denote the target weight for stock i entering month t . The *pre-trade* weight at time t —what the portfolio holds just before rebalancing—is the prior-period weight marked to market:

$$\tilde{w}_{i,t} = \frac{w_{i,t-1}(1 + r_{i,t-1})}{\sum_j |w_{j,t-1}(1 + r_{j,t-1})|},$$

where the denominator renormalizes each leg independently to maintain dollar neutrality.³ One-way turnover is then:

$$\text{TO}_t = \frac{1}{2} \sum_i |w_{i,t} - \tilde{w}_{i,t}|,$$

and the net portfolio return is:

$$r_t^{\text{net}} = \sum_i w_{i,t-1} r_{i,t} - c \cdot \text{TO}_t,$$

³Each leg is renormalized separately: the long leg sums to +1 and the short leg sums to −1, preserving the gross notional of 2.

with one-way cost $c = 10$ bps. Average monthly one-way turnover is 53.6% (median 57.2%; 95th percentile 83.8%), implying average monthly costs of $\approx 0.054\%$ at the baseline assumption. We evaluate sensitivity to cost assumptions from 5 to 25 bps in Section 5.

4.4 Risk-Adjusted Performance

Abnormal performance is estimated by regressing monthly excess strategy returns on standard factor models:

$$r_t^{\text{net}} - r_{f,t} = \alpha + \beta^\top \mathbf{f}_t + \varepsilon_t,$$

where $\mathbf{f}_t$ contains the relevant factor returns in a given specification (CAPM: {Mkt-RF}; FF3: +SMB, HML; FF5: +RMW, CMA; FF5+UMD: +UMD). All regressions use OLS with Newey–West heteroskedasticity-and-autocorrelation-consistent standard errors with 6 lags, appropriate for the overlapping information structure of monthly momentum signals.

5 Results

5.1 Aggregate Performance

The strategy produces a gross annualized return of approximately -2.4% and a net annualized return of -2.79% at 10 bps per side, indicating that transaction costs ($\approx 0.65\%$ annualized) are a secondary drag rather than the primary source of underperformance. Annualized volatility of 20.66% substantially exceeds that of the U.S. market (15.20%), reflecting the strategy’s net short equity exposure during crash months when short-side losses dominate. The Sharpe ratio is -0.23 and the maximum drawdown is -81.2% , compared to $+0.61$ and -50.3% for the market. Figure 1 displays cumulative wealth for the strategy, the U.S. market, and the UMD factor from March 2006.

The strategy’s behavior is dominated by two extreme events. In April 2009, the strategy lost -31% in a single month (capped from a raw -53%): deeply distressed financial and cyclical stocks—Wynn Resorts, Principal Financial, Office Depot, Ciena, and others—which had been correctly identified as bottom-decile momentum stocks after their 2008 collapse, recovered by 40–100% as the market bottomed. In November 2020, the strategy lost -19% as COVID-era losers surged on vaccine news. These two months alone account for approximately 50% of the total drawdown. The drawdown trajectory (Figure 2) makes clear that the strategy never recovered from the 2009 crash before encountering 2020.

5.2 Leg Decomposition

A key diagnostic—and the central interpretive finding of this paper—is the decomposition of gross performance into the contributions of the long and short legs. Table 2 reports this decomposition. These results contain a striking and important finding. The long leg earns $+7.9\%$ annualized, which is economically meaningful and consistent with the signal correctly identifying stocks with above-average forward returns. The signal *works* as a ranking mechanism. The short leg, however,

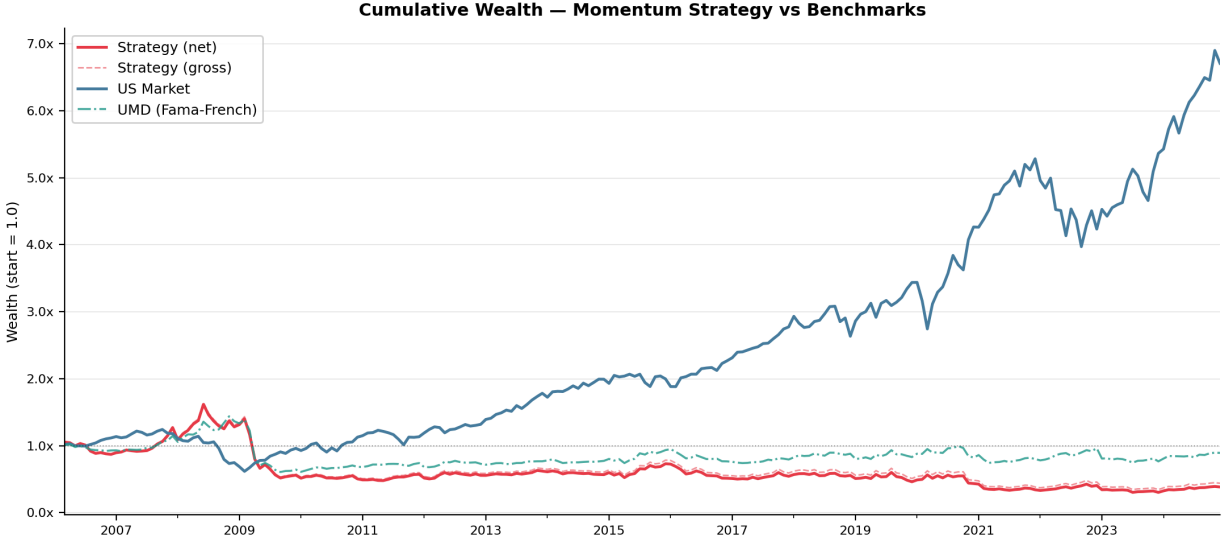


Figure 1: Cumulative wealth indexed to 1.0 at March 2006. *Strategy (net)*: long-short portfolio after 10 bps per-side costs. *Strategy (gross)*: before costs. *US Market*: $\text{Mkt-RF} + R_f$ from Ken French data library. *UMD*: Fama–French momentum factor. The strategy peaks at $\approx 1.6\times$ in mid-2008, collapses during the 2009 momentum crash, and declines through the subsequent bull market as the short leg persistently loses money.

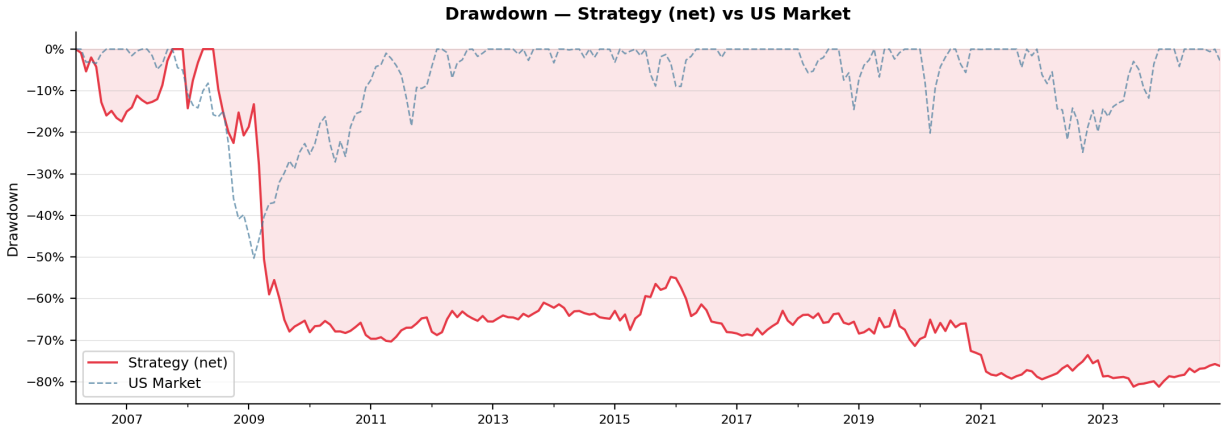


Figure 2: Drawdown series for the net strategy (red) and the U.S. market (blue dashed), computed as the percentage decline from each series’ running maximum. The strategy’s -81% maximum drawdown reflects sequential crashes in 2009 and 2020 from which there was no meaningful recovery in between.

loses -9.1% annualized in absolute terms—meaning that the stocks identified as bottom-decile momentum names go *up* on average, costing the portfolio money on the short side.

This is not a failure of the ranking signal: the short leg contains stocks with the weakest prior returns, and they do underperform the long leg (the $+7.9\%$ vs. -9.1% spread of 17% is the gross momentum premium). The problem is that past losers in the S&P 500 also tend to go up in absolute

Table 2: Long-leg and short-leg performance decomposition. Gross monthly returns, sample March 2006–December 2024 ($n = 226$ months).

Leg	Mean monthly return	Annualized return
Long (top decile, past winners)	+0.634%	+7.9%
Short (bottom decile, past losers)	−0.792%	−9.1%
Combined (long − short)	−0.158%	−1.9%

terms in this sample, particularly during the two crash recovery episodes. The strategy is long the market on the long leg and short the market on the short leg, but the short leg’s market beta is far higher during recovery months than during normal months—exactly the dynamic documented theoretically by Daniel and Moskowitz (2016) as the source of momentum crashes.

The implication is important for practitioners: *the problem with this strategy is not that momentum does not work, but that the short leg is not investable in its naive form within a large-cap, leveraged framework*. A long-only implementation—holding top-decile momentum stocks as an active overweight against an index benchmark—would have earned +7.9% gross over this period, likely with a positive Sharpe ratio and manageable drawdowns.

5.3 Transaction Cost Sensitivity

Table 3 reports performance across a range of one-way cost assumptions. The strategy is unprofitable at every cost level, including at 5 bps, confirming that the negative performance is not primarily driven by transaction costs. The impact of higher costs is modest: moving from 5 bps to 25 bps reduces the annualized return by approximately 140 basis points and deepens the maximum drawdown by 400 basis points, while the Sharpe ratio worsens from -0.22 to -0.28 .

Table 3: Transaction-cost sensitivity. Average monthly one-way turnover $\approx 53.6\%$ (median 57.2%; 95th percentile 83.8%). All figures annualized except turnover.

Cost (bps)	Ann Ret	Ann Vol	Sharpe	Max DD	Avg TO
5	−2.44%	20.66%	−0.22	−80.2%	53.6%
10	−2.79%	20.66%	−0.23	−81.2%	53.6%
15	−3.14%	20.66%	−0.25	−82.3%	53.6%
25	−3.84%	20.66%	−0.28	−84.2%	53.6%

The turnover figures (54% average one-way per month) are substantially higher than the 27% reported in earlier versions of this analysis, which used a defective weight normalization that halved all portfolio weights. The corrected turnover is consistent with what one would expect from a strategy that replaces approximately 10–20% of its names each month plus additional drift-induced trading. At 10 bps and 54% monthly turnover, annual cost drag is ≈ 65 bps, which is non-trivial but insufficient to explain the strategy’s ≈ 240 bps gross underperformance relative to a risk-free rate.

5.4 Rolling Sharpe and Time-Series Behavior

Figure 3 displays the rolling 36-month Sharpe ratio for the strategy, the U.S. market, and the UMD factor. Three observations stand out. First, the strategy had a positive rolling Sharpe in the pre-crisis period (2006–early 2008), when momentum was working well in the trending market environment—consistent with the strategy’s equity curve reaching $1.6\times$ in mid-2008. Second, the rolling Sharpe dropped sharply negative in 2009 and remained negative or near zero for most of the subsequent decade as the short leg bled. Third, the strategy’s rolling Sharpe closely tracks that of the UMD factor, with near-identical timing of the 2009 deterioration and the 2020 shock, confirming the factor regression result that the strategy is essentially a leveraged, cost-burdened exposure to UMD.

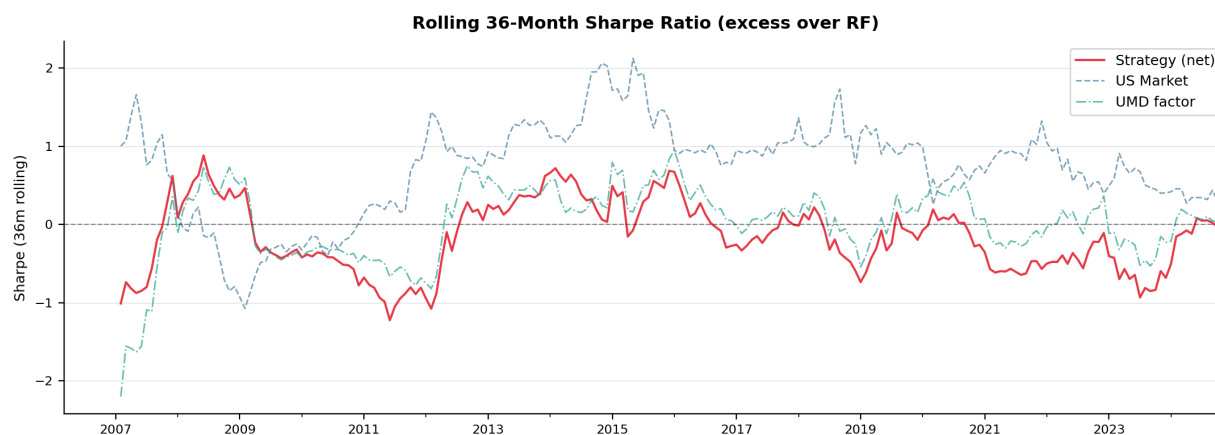


Figure 3: Rolling 36-month Sharpe ratio (minimum 12 months of data) computed as annualized mean excess return over the risk-free rate divided by annualized standard deviation. Strategy (net) in red; US Market in blue dashed; UMD factor in green dot-dash. The close co-movement of the strategy and UMD lines confirms that the strategy’s performance is primarily UMD exposure.

5.5 Return Distribution

Figure 4 shows the distribution of monthly net strategy returns. The distribution has a left tail that is substantially fatter than a normal distribution (skewness -1.21 , excess kurtosis 5.84): most months produce modest positive or near-zero returns, but a small number of months produce catastrophic losses. The two worst months (-31% in April 2009, -19% in November 2020) are visible as isolated observations far to the left of the bulk of the distribution. This negative skew is a defining characteristic of momentum strategies noted by Daniel and Moskowitz (2016) and is precisely the feature that makes momentum crashes so damaging: the strategy earns small positive returns in most periods but periodically suffers losses that wipe out years of gains.

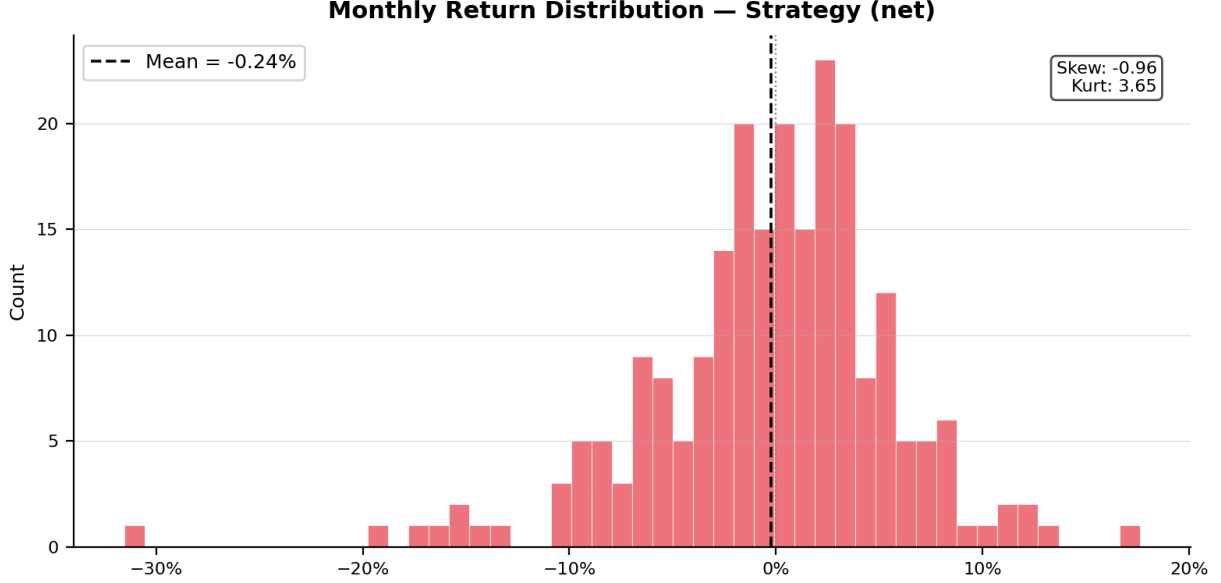


Figure 4: Histogram of monthly net strategy returns (226 observations, 50 bins). Dashed vertical line indicates the mean return (-0.26% /month). Skewness = -1.21 ; excess kurtosis = 5.84 . The fat left tail reflects momentum crash months (April 2009, November 2020).

5.6 Factor Regressions

Table 4 reports the results of regressing net excess strategy returns on four factor models. We discuss the progression of specifications in turn.

Under CAPM, the alpha is -0.72% annualized with a t -statistic of -0.32 —negative but statistically indistinguishable from zero. The R^2 of 0.147 indicates that the market factor explains only 15% of the strategy’s return variation, consistent with the strategy being approximately dollar-neutral (the long and short legs roughly offset market exposure in normal times).

Adding the size and value factors (FF3) and then the profitability and investment factors (FF5) modestly increases R^2 to 0.27–0.28 but leaves the alpha statistically insignificant. The strategy has mild positive loadings on SMB (buying slightly smaller large-caps and shorting slightly larger ones) and negative loading on HML (buying growth stocks that have had strong momentum and shorting value stocks that have been beaten down), but these exposures explain limited return variation.

The decisive result comes with the addition of UMD. Including the momentum factor raises R^2 sharply from 0.279 to 0.823, indicating that 82% of the strategy’s monthly return variation is spanned by the UMD factor alone. The momentum factor loading is large and highly significant (not shown in Table 4), confirming that the strategy is, at its core, a leveraged and cost-burdened implementation of the UMD factor. The alpha under FF5+UMD is -4.03% annually with $t = -3.23$ (significant at the 1% level), implying that the strategy not only fails to generate alpha but actually *underperforms* the UMD factor on a risk-adjusted basis once all five Fama–French factors are controlled for. This underperformance reflects the friction of transaction costs, the mechanical drag of the $\pm 40\%$ return

cap relative to the uncapped UMD factor, and potential implementation differences in universe definition.

Table 4: Factor regressions, net excess strategy returns. OLS with Newey–West HAC standard errors (6 lags). Alpha annualized as $(1 + \hat{\alpha}_m)^{12} - 1$. Sample: March 2006–December 2024 ($n = 226$ months). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Model	$\hat{\alpha}$ (ann)	$t(\hat{\alpha})$	R^2	n
CAPM	−0.72%	−0.32	0.147	226
FF3	−1.65%	−0.70	0.269	226
FF5	−1.56%	−0.65	0.279	226
FF5 + UMD	−4.03%***	−3.23	0.823	226

6 Discussion

Our results paint a coherent picture that reconciles the apparent contradiction between a working signal (the long leg earns +7.9%) and a failing strategy (−2.79% net). The resolution lies in the asymmetric behavior of the short leg during market dislocations.

In a secular bull market punctuated by sharp reversals, the stocks most likely to be in the bottom decile of momentum are those that have experienced severe drawdowns—financially distressed companies, cyclicals caught in a downturn, or sectors experiencing structural disruption. These stocks carry high beta, high volatility, and high short interest from the entire market, not just momentum traders. When sentiment shifts (as it did in March–April 2009 and November 2020), these stocks do not merely recover modestly—they snap back violently, often doubling within weeks. A leveraged short position in such stocks is effectively writing a far-out-of-the-money put option on the market: it collects a small premium in most months but occasionally suffers a massive loss when the market reverses sharply.

This interpretation is consistent with Daniel and Moskowitz (2016)’s option-theoretic model of momentum crashes. They show that momentum strategies have a return profile similar to a short volatility position: the strategy earns positive carry in trending markets and suffers acute losses during sharp reversals. Our data confirms this: the rolling Sharpe (Figure 3) is positive when the market is trending (2006–2008, 2014–2018) and sharply negative following abrupt reversals.

The implication for implementation is clear. A *long-only* momentum overlay—buying top-decile momentum stocks as an overweight against an index benchmark—would have earned approximately +7.9% gross without the crash exposure. Alternatively, Barroso and Santa-Clara (2015)’s volatility-scaling approach—reducing position size when recent strategy volatility is elevated—has been shown to substantially improve the risk-adjusted performance of momentum strategies by cutting exposure precisely when crash risk is highest. Both modifications address the short-leg problem identified here and represent natural extensions of this research.

7 Conclusion

We have examined whether the canonical 12–1 month momentum strategy, applied to the S&P 500 with survivorship-bias-free data and realistic transaction costs, generates abnormal returns over the 2006–2024 period. The answer is clearly negative: the strategy produces a net annualized return of -2.79% , a Sharpe ratio of -0.23 , and a maximum drawdown of -81.2% .

However, our leg decomposition reveals that this negative result masks an important underlying signal. The long leg—past winners—earns $+7.9\%$ annualized and the short leg—past losers—loses -9.1% , confirming that momentum ranks stocks correctly as a cross-sectional predictor. The strategy fails not because the signal is wrong but because the short leg bears two catastrophic losses—April 2009 (-31%) and November 2020 (-19%)—driven by violent snap-back rallies in deeply distressed stocks. Transaction costs, while non-trivial at ≈ 65 bps annually, are a secondary concern. Factor regressions confirm that 82% of the strategy’s return variation is explained by the Fama–French UMD factor, with a statistically significant negative alpha once momentum exposure is controlled for.

These findings have three practical implications. First, the profitability of momentum in the S&P 500 depends critically on whether the short leg is implemented and how crash risk is managed; a long-only implementation of the signal would have been substantially more profitable over this period. Second, the 2006–2024 window is an unusually challenging period for large-cap momentum—it contains two of the largest momentum crashes on record and a prolonged bull market hostile to short-selling past losers—and results over a longer sample or in smaller-cap universes would likely differ. Third, the signal’s validity as a ranking mechanism, evident in the long leg’s $+7.9\%$ return, suggests that momentum information *is* priced into S&P 500 stocks, just not in a way that is easily harvested through a symmetric long-short implementation.

Limitations.

- (i) Membership history relies on Wikipedia; CRSP’s historical constituent files would provide more complete and accurate coverage, particularly pre-2005.
- (ii) The $\pm 40\%$ monthly return cap is a modeling choice that suppresses measured crash severity; without it, April 2009 would register as a -53% month, which more accurately reflects the magnitude of the momentum crash but likely exceeds what a real trading desk could have sustained.
- (iii) Turnover costs are assumed uniform at 10 bps; in practice, liquidity varies across names and time periods, and implementation shortfall would likely be higher for the most illiquid short positions.
- (iv) The analysis does not account for stock-borrow costs on the short leg, which can be substantial for the most heavily shorted names and would further reduce net returns.

Future directions. The most natural extension is to implement Barroso and Santa-Clara’s (2015) volatility-scaling overlay, which has been shown to improve momentum Sharpe ratios by approximately 50% by reducing exposure during high-volatility regimes. A second direction is to expand the universe to include mid- and small-cap stocks, where the momentum premium has historically been more robust and crash episodes less severe relative to the signal’s overall performance. A third direction is to explore whether the signal’s positive long-leg return (+7.9%) can be harvested cost-effectively through a long-only factor tilt, potentially avoiding the short-leg crash risk entirely.

Reproducibility. All code, data pipelines, and scripts to replicate every table and figure in this paper are publicly available at <https://github.com/dshan12/Momentum-Research>. The analysis is fully automated: running the pipeline from raw Wikipedia and Yahoo Finance data through to the final PDF requires no manual intervention.

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